

Simple X and O Game

Python Program

```
# Simple X and O Game

board = [" " for _ in range(9)]

def show():
    print(board[0], "|", board[1], "|", board[2])
    print("---+---+---")
    print(board[3], "|", board[4], "|", board[5])
    print("---+---+---")
    print(board[6], "|", board[7], "|", board[8])

def win(player):
    combinations = [
        (0, 1, 2), (3, 4, 5), (6, 7, 8),
        (0, 3, 6), (1, 4, 7), (2, 5, 8),
        (0, 4, 8), (2, 4, 6)
    ]

    for a, b, c in combinations:
        if board[a] == board[b] == board[c] == player:
            return True
    return False

player = "X"
moves = 0

while moves < 9:
    show()

    try:
        pos = int(input(f"Player {player}, enter a position (1-9): ")) - 1
    except ValueError:
        print("Please enter a number from 1 to 9.")
        continue

    if pos < 0 or pos >= 9:
        print("Position must be from 1 to 9.")
        continue

    if board[pos] != " ":
        print("Position already taken!")
        continue

    board[pos] = player
    moves += 1

    if win(player):
        show()
        print(f"Player {player} wins!")
        break

    player = "O" if player == "X" else "X"
else:
    show()
    print("Game Draw!")
```